

with family incomes above the poverty line starting in 2014 in non-expansion states.

Finally, it is important to note that the insurance coverage changes documented in the ACS and NHIS only provide information on coverage changes in any given year. However, just as the ACA Medicaid expansions affected the accumulation of Medicaid coverage over time, they also affected the accumulation of insurance coverage. We do not have longitudinal administrative data on the total number of years of insurance coverage as we do with Medicaid enrollment. However, we are able to shed light on cumulative insurance coverage using data from the HRS.²¹ Our sample in the HRS is much smaller than the ACS-mortality panel (N=1,359 vs. 566,000) but allows us to provide at least some suggestive evidence regarding cumulative insurance coverage. We find that respondents living in expansion states report having accumulated 0.39 additional years of continuous insurance coverage in the post-expansion period relative to respondents in the non-expansion states. This estimate is imprecise and only significant at the 10 percent level, albeit of a similar magnitude to the cumulative change in Medicaid coverage that we document in both the HRS (Appendix Table A3) and the administrative CMS data (Table I).

Taken together, our first stage analysis indicates that there were large and significant impacts of the ACA Medicaid expansions on eligibility, Medicaid enrollment, and insurance coverage for our target sample. In addition, individuals accumulated more years of exposure to Medicaid and insurance coverage as a result of these expansions.

V.B Mortality

The previous section established that the Medicaid expansions had a meaningful impact on eligibility, enrollment, and coverage for our sample. We now examine the impact of this expansion on mortality. Our estimates of equation (1) are presented in Figure II and in the sixth column of Table I. Prior to the ACA expansion, mortality rates trended similar across the two groups: pre-expansion event study coefficients are close to zero and not statistically significant. Starting in the first year of the expansion (year 0), we observe mortality rates decrease significantly among respondents in expansion states relative to non-expansion states. The coefficient estimated in the first year of expansion indicates that the probability of annual mortality declined by 0.089 percentage points. In years 1 and 2, we find reductions in the probability of a little over 0.1 percentage points and, in year 3, a reduction of 0.208 percentage points. All estimates are statistically significant, and our confidence intervals include meaningfully-sized effects.²²

The immediate impact of the Medicaid expansion on mortality mirrors several analyses that find sizable changes in health care utilization during the first year of the expansions. For example, Garthwaite et al. (2017) find large changes in hospital ED usage; Wherry and Miller (2016) find increases in hospitalizations, physician visits, and diagnoses of chronic illnesses; Ghosh et al. (2019) find increases in the use of prescription drugs; and, Sommers et al. (2015) find improvements in access to medication and personal physicians all emerging within the first year of the Medicaid expansions.

In addition, we find evidence that the mortality effects of the expansions are growing over time. As discussed earlier, mortality is increasing over time for our panel as they reach older ages. However,

²¹See Appendix Section 3 for details on the HRS sample construction and analysis.

²²For example, in year 0, we are able to rule out reductions in mortality smaller in magnitude than 0.018 percentage points; in year 1, 0.033 percentage points; in year 2, 0.021 percentage points and in year 3, 0.045 percentage points.

when we examine the coefficient estimates as compared to a counterfactual mortality rate for each event time, we find that the proportionate change in mortality is increasing in each year, from an estimated 7.0 percent reduction in year 0 to a 11.9 percent reduction in year 3.²³

The top panel of Table I shows the difference-in-differences estimate that pools all post-expansion years together. Using this model, we estimate an average reduction in annual mortality of 0.132 percentage points. This represents a reduction in mortality of about 9.4 percent relative to the sample mean, or 8.1 percent relative to our estimated counterfactual mortality rate during the post period of 1.63 percent.²⁴

We conduct additional analyses to identify the changes in death rates by the underlying cause of death using the MDAC. These analyses rely on a much smaller sample and shorter follow-up period, and so we consider this analysis to be exploratory in nature. We examine deaths due to non-disease related (i.e. “external”) and disease-related (i.e. “internal”) causes separately. We also examine a subset of deaths caused by internal factors that are considered to be “health care amenable” (Nolte and McKee, 2003), which have been studied in the existing literature (e.g. Sommers et al., 2014; Sommers, 2017).

These results are presented in Table II. We observe similar patterns for internal mortality and health care amenable mortality as we do in our main results, with relative decreases beginning in the first year after the expansions occur. The event study coefficients are not statistically significant for health care amenable mortality, and are significant at the $p < 0.10$ level for deaths from internal causes; however, we find a highly significant reduction in deaths related to internal causes under the difference-in-differences model. In contrast, mortality from external causes, which may be less affected by insurance coverage, does not appear to decrease after the expansions. The point estimates on the event study indicators are not statistically significant and the difference-in-differences estimate is only significant at the 10 percent level. The estimate is positive in sign, although we note that there is a slight upward pre-trend in these deaths in expansion states relative to non-expansion states.

We further probe cause of death by conducting an analysis using the ICD code groupings by body system. This exercise is meant to be exploratory with the hope that it will provide guidance for future work should better data become available. The results are reported in Appendix Table A4. For most diseases, we observe negative coefficients; the largest negative point estimates are observed for deaths related to neoplasms (cancer), endocrine and metabolic diseases (primarily diabetes), cardiovascular and circulatory system diseases, and respiratory diseases. Two of these (cardiovascular/circulatory and endocrine/metabolic) are marginally significant at the 10 percent level. The point estimates suggest that cardiovascular disease might account for approximately 38 percent of the overall reduction in internal mortality, with endocrine and metabolic diseases accounting for another 18 percent. Changes in mortality for these two body groupings are consistent with recent evidence documenting a decrease

²³As described earlier, we estimate a counterfactual mortality rate that takes into account both that our panel is aging over time and the mortality reduction of expanded Medicaid coverage during the post-period. For each event time, it is equal to the sum of the post-expansion mortality rate in expansion states in that period and the absolute value of the corresponding event study coefficient for that period. The counterfactual rates are 1.28 percent in year 0, 1.51 percent in year 1, 1.60 percent in year 2, and 1.75 percent in year 3. We are then able to express our reduced form effects as a 7.0 percent reduction in year 0, 7.9 percent reduction in year 1, 8.2 percent reduction in year 2, and 11.9 percent reduction in year 3.

²⁴The counterfactual mortality rate is calculated as the average annual mortality rate observed in the in expansion states in the post-expansion period (1.494 percent) minus the reduced form mortality difference-in-differences estimate (-0.1320pp).

in metabolic syndrome related conditions (obesity, high blood pressure, diabetes) and complications arising from these conditions among near-elderly adults under the Medicaid expansions (McInerney et al., 2020). We also see a small negative but statistically significant impact on diseases related to the skin and subcutaneous tissue, for which diabetes is an important risk factor (Ki and Rotstein, 2008).

VI Alternative Specifications and Robustness Checks

VI.A *Role of Staggered Treatment Timing*

Recent research demonstrates that variation in treatment effects across different “treatment cohorts,” defined by their treatment timing, can make event study estimates difficult to interpret. In our context, we have four different treatment cohorts with 21 states expanding Medicaid in 2014, 3 states expanding in 2015, 2 states in 2016, and 1 state in 2017. Sun and Abraham (2020) show that, if each of these cohorts have different profiles of time-varying treatment effects, the event study estimates can be contaminated by treatment effects from other periods. We demonstrate that this variation in treatment timing is not a concern in our context by conducting additional analyses that (1) limit the expansion states to the 2014 expanders, and (2) use the alternative estimation method proposed in Sun and Abraham (2020). The event study results from these two approaches are reported in Appendix Figure A2 and are very similar to our main results. In addition, the difference-in-differences estimate when dropping the late expanding states is reported in the second row of Figure III, with the estimate from our main specification provided in the first row for comparison purposes. The estimate is very similar to our main result. Further details on these analyses may be found in Appendix Section 5.

In related work, Goodman-Bacon (2019) demonstrates that staggered treatment timing combined with the presence of time-varying treatment effects can lead to biased DD estimates. This can occur when earlier treatment cohorts function as controls for later treatment cohorts while on a differential path due to their earlier treatment. We do not expect this to be a major concern in our context given that we have so few late adopter states and a relatively short post period. However, to determine the potential influence of comparisons of states with different treatment times, we implement the Goodman-Bacon (2019) decomposition that examines the role of each 2x2 DD comparison in the two-way fixed effects DD estimate.²⁵ We find that only 11 percent of the DD estimate is derived from comparisons of states with different treatment times. The DD estimate that excludes this source of variation, relying solely on comparisons of treated and untreated units, is very similar, although slightly larger in magnitude than our main DD estimate (see Table A5).

VI.B *Further Investigating Differential Pre-Trends*

Next, we consider the potential impact of differential linear pre-trends in expansion and non-expansions states. While we do not find any evidence of differential pre-trends in the event study, we explore how well-powered this test is in our context. To do this, we first determine the size of a linear pre-trend we are powered to detect following the procedure described in Roth (2019). Our analysis suggests that we could detect with 80 percent power a fairly small negative linear trend of a magnitude of 0.03235 percentage points or greater (in absolute terms) in our event study model (i.e. a pre-trend of such size

²⁵We implement this decomposition following the methods outlined in Goodman-Bacon (2019) and based on the code in Goodman-Bacon et al. (2019).

is likely to generate at least one statistically significant pre-period event study coefficient). If a trend of a size of up to -0.03235 percentage points is indeed present (although not detectable to us), we calculate that it would generate, by year 3 following the expansion, a bias of at most -0.08873 percentage points.²⁶ Our actual estimate in year 3 is -0.2082 percentage points, and substantially larger (2.3x) than this potential bias.

Next, we explicitly allow for differential linear trends in the model. First, we estimate an alternative version of the model that allows for differential pre-trends in expansion and non-expansion states by replacing the pre-expansion event study coefficients with a linear trend in event time for expansion states, as in, e.g. Dobkin et al. (2018) and Gross et al. (2020). The estimates of this specification are reported in Appendix Table A6 and are very similar to our main estimates. Second, we estimate a version of the model that includes state-specific linear pre-trends, following the two-step procedure proposed by Goodman-Bacon (2019). The result from this approach is presented in the third row of Figure III and is similar to our main model. Third, we estimate a model that controls for local trends in mortality among counties with different demographic and economic characteristics by interacting the 2013 county-level unemployment rate, median income, poverty rate, share Black, share Hispanic, and share female population with linear year trends. Including these trend terms does not appreciably affect our estimate, as seen in the fourth row of Figure III.²⁷

VI.C *Confounding Factors and Threats to Validity*

Even if outcomes were evolving similarly for the two groups of states prior to expansion, the assumptions of our model would be violated if they experienced differential economic or policy shocks around the time of the expansion that drove changes in health outcomes. We explore this by introducing different sets of covariates and examining the impact of their inclusion on our estimates. The difference-in-differences estimates are presented graphically in Figure III. We also include event study versions for these alternative specifications, with our original estimates depicted in grey for comparison, in Appendix Figure A3.

We first examine sensitivity to controls for local economic conditions. In the fifth row of Figure III, we control for predicted changes in labor demand at the county level (see Appendix Section 4 for further details). Row 6 includes time-varying controls for county-level economic characteristics (unemployment rate, poverty rate, median household income). Our estimate remains unchanged with the inclusion of these variables.

We next examine whether differential exposure to the opioid crisis could be an important confounding factor. Analysis of the MDAC data shows that the change in mortality we observe is driven by a decrease in deaths due to internal, disease-related causes, which do not include drug overdoses. However, we further explore this possibility by directly controlling for state policies that have been tied to opioid overdose rates in the seventh row of Figure III. This model includes the following controls: indicators for prescription drug monitoring programs (PDMP), mandatory PDMPs that require physicians to access patients' prescription histories, state regulations for pain clinics, legalization of medical

²⁶We calculate the bias following the formula presented in Roth (2019), which takes into account the additional bias introduced by passing a pre-test.

²⁷We also consider the presence of non-linear differential pre-trends using methods outlined in Rambachan and Roth (2019). Our analysis suggests our results are reasonably robust to unrelated deviations in the two groups. See Appendix Section 4 for more details.